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IARCO RESEARCH PROPOSAL

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Research Topic: Oxidized Metal Surface Disinfection: An Analysis for a Correlation to Basic Metal Properties

Title: Oxidized Metal Surface Disinfection: An Analysis for a Correlation to Basic Metal Properties

Research Problem

The COVID-19 pandemic has brought attention to microorganisms that can mingle on commonly touched surfaces in areas like hospitals, medical centers, public transportation, local markets and industrial zones [1]. While the U.S. Environmental Protection Agency (EPA) has confirmed that HVAC filters can help reduce airborne contaminants [2], surfaces remain a problem. Considering that in previous years multiple studies have shown that fewer than half of the hospital touch surfaces are adequately cleaned [3, 4], and among the microorganisms that were found on contaminated surfaces were antibiotic resistant microorganisms [5], it is crucial to find an efficient self-disinfecting mechanism. It is especially critical to find ways to slow the spread of microorganisms due to the massive uses of antibiotics creating a surge in antibiotic resistant microorganisms, which has attracted the attention of the Defense Advanced Research Project Agency (DARPA) and the Advanced Research Projects Agency for Health (APRA-H), who have launched and funded multiple projects with the goal of combating antibiotic resistant microorganisms [6-10].

Scientists have been looking into using metals as self-disinfecting surfaces to prevent the spread of microorganisms from inadequately cleaned surfaces since the 20th century, with a sharp increase of research in the field after the COVID-19 pandemic. However, data and results reported in publications has one significant gap: the absence of comparative studies that involve a range of oxidized metals. This prevents establishing fundamental trends and properties of the disinfecting action of the oxidized metal surfaces, leaving room for potential correlation to a pre-established periodic trend. Determining this periodic trend would help scientists better understand bactericidal mechanisms. This research aims to address these gaps in knowledge, with a focus on metal oxide interactions with a gram-negative bacteria, *Escherichia coli* K-12 strand, and a gram-positive bacteria, *Micrococcus luteus*, comparing all to a control glass surface.

Existing Literature

When focusing on metal surface disinfection, previous studies have mainly focused on proving disinfection of metals, with a narrow focus on mainly copper and its alloys and their possible disinfection mechanism [11-33]. Silver [41-52] is the next most researched metal for disinfection. This creates a problem by possibly overlooking metals that may be more efficient at disinfection than copper and silver. Some research has involved studying unorthodox metal alloys [34-37]. This research contributes to bettering the understanding of disinfection of different metals. The most expansive study tested 9 different metal surfaces for their antibacterial properties and determined that not all metals are antibacterial [38]. Moreover, very few studies account for the factor of oxidation of metals [34, 39]. Lastly, scientists are still not fully sure about the antimicrobial mechanisms of metals [40].

This study attempts to fill gaps by firstly broadening the range of studied metals, secondly accounting for oxidation, and thirdly, trying to locate a correlation with a periodic trend for oxidized metal surface disinfection of bacteria.

Research Questions

1) If transition metals, most of which are essential for the proper function of living organisms, are taken, will they all have disinfecting properties?
2) What fundamental properties of metals dictate antimicrobial properties?
3) Will some of the metals be more efficient at disinfecting gram positive or gram negative bacteria?

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Methodology

This study involves a combination of chemical, biological, culturing and statistical approaches. The chemical, biological and culturing approaches are based on the works of Y. Miyano et al and S. A. Wilks et al.

Chemical component:

- Titanium, chromium, iron, nickel, copper, zinc, molybdenum, silver, tungsten and gold are studied, and compared to a control glass sample
- Each surface is cut into a 30 mm x 30 mm square and polished to a surface roughness of <0.25 micrometers, then put into 100 mL of glacial acetic acid (CH_3COOH) with glass beads on a lab shaker for 20 minutes, then drenched in 10 mL of acetone.
- The metal surfaces are then left to oxidize in open air in the lab away from direct sunlight and at humidity from 40% to 50% for 24 hours.
- Before culturing, each metal surface is dipped into 10 mL of 70% isopropyl alcohol for 1 minute and then put under UV light for 5 minutes

Biological component:

- Create an initial bacteria culture of *Escherichia coli* K-12 strand and a separate culture of *Micrococcus luteus* that has a CFU of at least 10^6 cells ml^-1
- Transfer the culture to a sterile 100 mL Schott bottle with 50 mL tryptone soy broth (TSB) with microbeads and store the culture at -80° C
- Before each culturing, thaw the Schott bottle at room temperature, take a single bead and aseptically transfer to a sterile 20 mL nutrient broth containing 5 mL TSB.
- Incubate the culture at 37° C for 12 hours.
- Do not use cultures that are over 20 hours old

Culturing component:

- 50 microliters of the prepared culture is deposited onto the precleaned surface and covered with a glass microscope cover slide, and a timer starts.

- Each surface is allowed to interact with bacteria for 1, 2, 3, 4, 5, 7, 8, 9, 10, 11, or 12 hours before continuing onto the wash procedure.
- In order to measure antibacterial activity of metals, wash the surface and the slide 3 times with a total of 15 mL of phosphate-buffered saline, collecting the phosphate-buffered saline with bacteria into a beaker
- 5 mL of the phosphate-buffered saline is cultured on tryptone soy agar (TBA) at 37° C for 24 hours, and colonies are counted after 24 hours to determine CFU
- Repeat the previous procedure for all 10 metals and the glass sample thrice for each bacteria type and each interaction time.
- In order to disinfect the agar growths, tested surfaces and containers that held bacteria, autoclave everything at 121° C for 30 minutes.

Statistical approaches:

- The mean of three data points for each metal, time and bacteria is calculated, and a 2-way ANOVA with repeated measures is used to prove or disprove if both time and surface have an effect on bacterial growth for the two different bacteria.
- The null hypotheses are rejected if the F-value is greater than the critical value that correlates with the α-value found on a statistics table, and if the p-value is less than 0.001.
- After conducting the 2-way ANOVA with repeated measures, if the null hypotheses are rejected for both time and different surfaces, then the next step can be taken. If the null hypotheses are accepted, then move on to writing a conclusion about the data
- If the null hypotheses are rejected, then create four graphs. Split the metals into normal and hybridized orbital groups, and create a total of four groups (two groups for the two different bacteria strands).
- Describe the general trends from one point to the next point of the graph, and compare this trend to preexisting periodic trends for transition metals, and identify the trend that the collected data follows the closest.

Expected Outcome

The study expects to find a correlation between the collected data and basic metal properties and their antimicrobial efficacy.

Benefit of the Study

The outcomes of the study create a foundation for predictive capabilities for the field of metal surface disinfection. Expected trends hint at novel efficient unorthodox antimicrobial surfaces. Additionally, the obtained data will provide a new perspective on disinfecting mechanisms of metal surfaces.

Potential Limitation

Because the data requires a grand total of 792 cultures, there may be discrepancies in the data. Additionally, there may be some variation in the oxidation state of the metals due to differences in humidity, temperature and sunlight exposure in the lab. The bacteria batches will need to be

remade as they will run out, which may create discrepancies in the data. Lastly, human errors may influence the results.

Ethical Consideration

All researchers are to wear personal protection equipment such as single-use gloves, goggles and lab coats or aprons.
All microorganisms that are being worked with in the lab are to be treated as pathogens, and all unknown cultures are to be treated as contagious pathogens and are autoclaved immediately. No culture is left for over 24 hours, and is autoclaved after counting colonies. All equipment that is being worked with in the microbiology lab is to be sterilized at 121 degrees Celsius for 30 minutes to prevent contamination.
The chemical waste generated from working with metals is to be properly disposed of. Glacial acetic acid is diluted to 10% and then poured down the drain. Metal acetates are taken to a specialized facility to be properly disposed of. When polishing the metals, the person doing so must wear a respirator and work under a fume hood.

Project Practicalities

This project will span over approximately 7 months. The first month will be spent acquiring the needed metals and bacteria. The next 5 months will be spent culturing bacteria in the lab and collecting data. The last month will be spent on data analysis, finalizing results and creating a full report.

Post-program Plan

Upon determining the periodic trend, experiments can be conducted to confirm that this periodic trend is the main antimicrobial mechanism

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